

HADAD: Multi-Functional Autonomous Robot

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Award: Recognized as one of the Top 2 Projects university-wide.

Overview

HADAD is a multi-functional autonomous robot designed from the ground up to merge advanced navigation with real-time AI. Engineered for robust environmental interaction, it features a dual-control architecture allowing for Global IoT operation via mobile and localized Gesture Control.

Core Features

Intelligence & Computer Vision

- **Object Detection:** Implemented a sophisticated intelligence suite using YOLO for object detection.
- **Feature Matching:** Utilizes SIFT for feature matching.
- **Dynamic Tracking:** Capable of dynamic environment tracking through optical flow, alongside people counting and image segmentation.

Navigation & Movement

- **Precision Movement:** Developed core logic based on Odometry, Kinematics, and Localization for high-precision movement.
- **Steering Mechanism:** Utilizes Differential Steering for agile maneuvering.

Control & Connectivity

- **IoT Architecture:** Engineered a global IoT control architecture using ESP32 and Arduino.
- **Interaction:** Allows for remote operation and seamless human-robot interaction.

Hardware & Mechanical Integrity

- **Chassis Build:** Designed and fabricated a heavy-duty chassis using iron and wood.
- **Suspension:** Features a custom vibration-absorbing suspension system for traversing uneven terrain.

Technology Stack

- **Software & AI:** Python, C++, OpenCV, YOLO, SIFT, TensorFlow, Optical Flow.
- **Hardware & IoT:** Arduino, ESP32, Raspberry Pi.
- **Robotics Math:** Odometry, Kinematics, Localization, Feature Matching.
- **Deployment & Tools:** Docker, FastAPI, Git, GitHub.

Potential Use Cases

Given its robust build and AI capabilities, HADAD is designed for:

- **Remote Facility Inspection:** Using global control to monitor hazardous or distant industrial sites.
- **Smart Security:** Utilizing people counting and object detection for automated patrolling.
- **Logistics & Inventory:** Navigating warehouses to track assets using image segmentation.
- **Search & Rescue:** Leveraging the vibration-absorbing suspension to traverse debris while streaming visual data.

Contact

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